DAMASCUS HEAT-AI

Project Idea: AI-Powered Urban Heat Exposure and Priority Mapping for Climate-Resilient Urban Planning in Damascus, Syria

1- **Problem & Intended Users:** Damascus faces increasing urban heat challenges due to dense urban development, heat-absorbing surfaces, and declining vegetation cover, contributing to the Urban Heat Island (UHI) effect. The lack of integrated spatial data linking land surface temperature (LST), urban characteristics, vegetation, and population distribution makes it difficult to identify areas requiring priority climate adaptation interventions.

DAMASCUS HEAT-AI aims to use satellite imagery and Artificial Intelligence (AI) to analyze Land Surface Temperature (LST), which represents the temperature of the Earth's surface rather than air temperature, and examine its relationship with vegetation, urban development, and population distribution.

The main goal is to identify areas where high land surface temperatures overlap with high population density and limited vegetation cover. The project will produce maps and spatial information to help relevant authorities prioritize urban greening, environmental interventions, and climate adaptation efforts in Damascus.

Intended Users: Municipalities, urban planners, environmental organizations, engineering firms, academic researchers, and local communities.

2- **SDG Pathway:** The project contributes to three Sustainable Development Goals:

-SDG 11 – Sustainable Cities and Communities: Supporting climate-resilient urban planning and identifying priority areas for green infrastructure.

-SDG 13 – Climate Action: Providing spatial evidence to support local climate adaptation strategies.

-SDG 3 – Good Health and Well-being: Identifying areas with potentially higher population exposure to surface heat to inform future risk mitigation efforts.

3- Two Relevant References

1. Khalil, M., & Kumar, J. S. (2025). Time-series studies of land surface temperature in Damascus, Syria through MODIS by Google Earth Engine. Advances in Space Research, 75(2), 1960–1973. https://doi.org/10.1016/j.asr.2024.10.044

This study provides a local scientific foundation by examining urban temperature patterns in Damascus and their relationship with vegetation and urban expansion.

2. Wang, X., Hsu, A., & Chakraborty, T. (2023). Citizen and machine learning-aided high-resolution mapping of urban heat exposure and stress. Environmental Research: Infrastructure and Sustainability, 3(3), 035003. https://doi.org/10.1088/2634-4505/acef57

This study demonstrates the potential of integrating machine learning and spatial data to produce high-resolution urban heat exposure maps.

Candidate Data: The project will use open-access geospatial datasets:

-Landsat 8/9: Land Surface Temperature (LST).

-Sentinel-2: Vegetation (NDVI) and built-up area (NDBI) indicators.

-WorldPop: Gridded population estimates for population exposure assessment.

-MODIS: Complementary data for broader temporal temperature analysis.

4- **Proposed Approach**: 1- Data Collection: Define the Damascus study area and collect suitable summer satellite imagery and population datasets.

2- Geospatial Analysis: Extract LST, NDVI, and NDBI indicators to examine relationships between urban characteristics and surface temperature.

3- AI-Based Analysis: Apply Random Forest to analyze the environmental and urban factors associated with heat patterns and classify areas by heat intensity.

4- Exposure Assessment: Integrate heat, vegetation, and population data to develop a Thermal Priority Index identifying areas requiring greater environmental intervention.

5- Visualization: Develop an interactive geospatial dashboard using Python and Streamlit to present heat patterns, population exposure estimates, and priority areas for urban greening.

Expected Output: An AI-supported urban heat assessment framework and interactive mapping dashboard providing actionable spatial insights for climate-resilient urban planning in Damascus.